

| Assessment | Test | Domain  | Skill            | Difficulty                               |
|------------|------|---------|------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard <div></div> <div></div> <div></div> |

Question

For the function  $f$ ,  $f(cx) = x - 8$  for all values of  $x$ , where  $c$  is a positive constant. If  $f(2) = 35$ , what is the value of  $c$ ?

Correct Answer: .0465, 2/43

Rationale

The correct answer is  $\frac{2}{43}$ . It's given that  $f(cx) = x - 8$  for all values of  $x$ , where  $c$  is a positive constant, and  $f(2) = 35$ . Therefore, for the given function  $f$ ,  $cx = 2$ . Dividing both sides of this equation by  $c$  yields  $x = \frac{2}{c}$ . Substituting  $\frac{2}{c}$  for  $x$  in the equation  $f(cx) = x - 8$  yields  $f(\frac{2c}{c}) = \frac{2}{c} - 8$ , or  $f(2) = \frac{2}{c} - 8$ . Since it's given that  $f(2) = 35$ , substituting 35 for  $f(2)$  yields  $35 = \frac{2}{c} - 8$ . Adding 8 to both sides of this equation yields  $43 = \frac{2}{c}$ . Multiplying both sides of this equation by  $c$  yields  $43c = 2$ . Dividing both sides of this equation by 43 yields  $c = \frac{2}{43}$ . Note that 2/43, .0465, 0.046, and 0.047 are examples of ways to enter a correct answer.